



DATA WITH BARAA

WINDOW FUNCTIONS

BASICS

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YouTube | **DATA WITH BARAA**
SQL Course | Window Functions



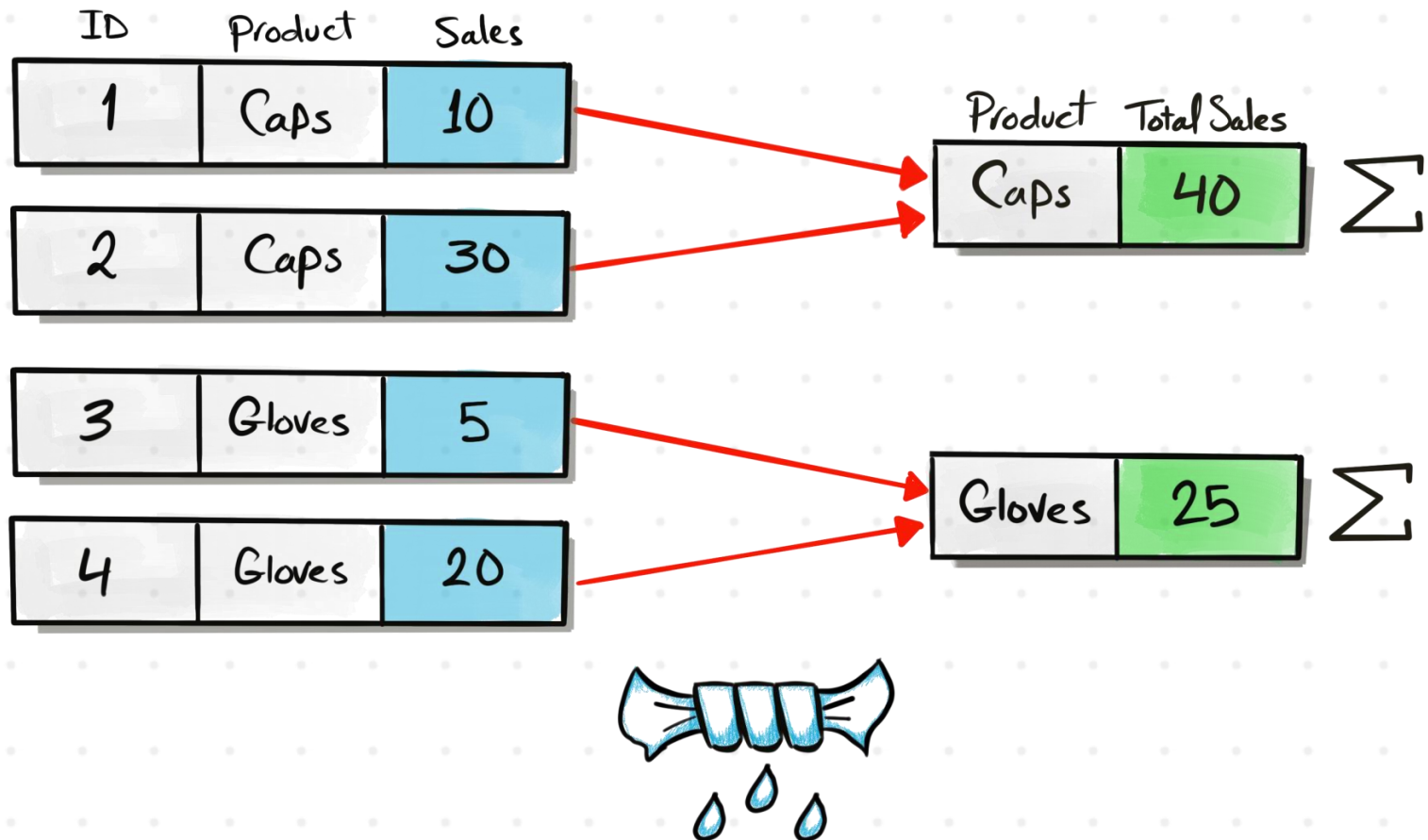


WINDOW FUNCTIONS

**Perform calculations (e.g. aggregation)
on a specific subset of data,
without losing the level of details of rows.**

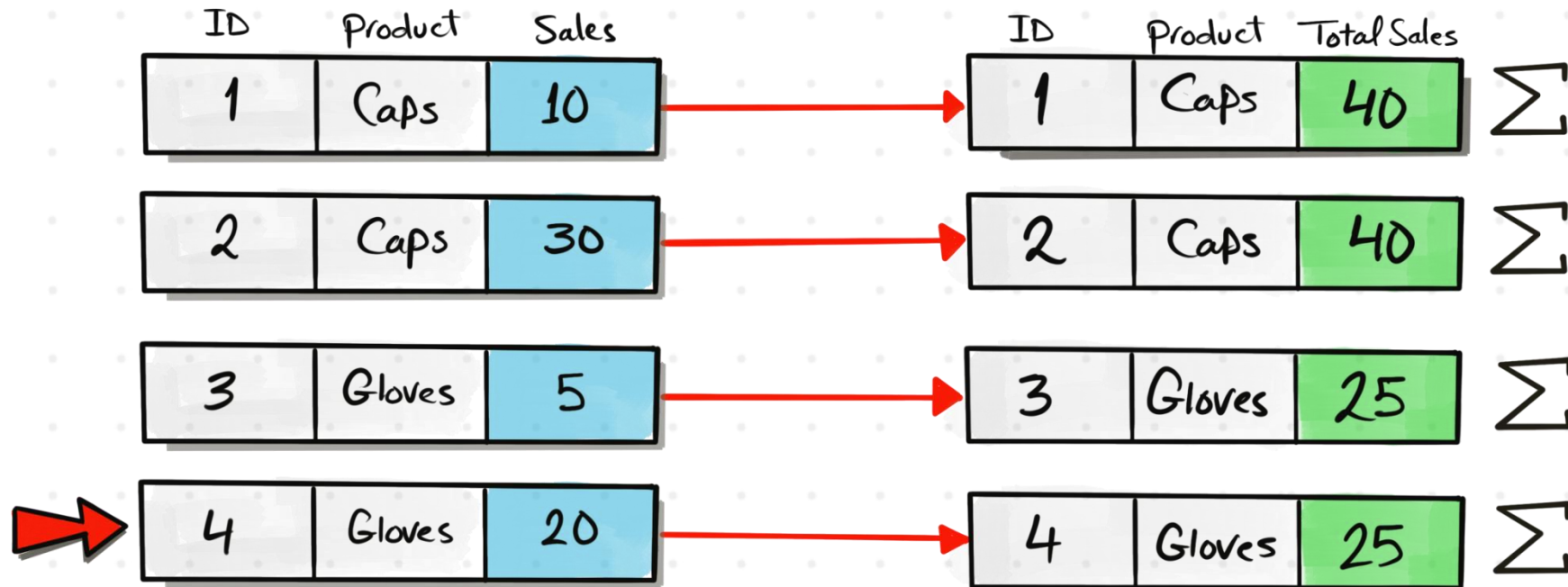
GROUP BY

Aggregates and groups rows **based** on column/s into **summary rows**



WINDOW Functions

Compute **aggregates** but **keep details** of individual rows at the same time



Row-Level Calculation

GROUP BY

ID	Product	Sales	
1	Caps	10	
2	Caps	30	
3	Gloves	5	
4	Gloves	20	

Product	Total Sales	
Caps	40	Σ
Gloves	25	Σ

It **collapse** groups of rows into a single row
(Group-Level-Calculations)

Window Functions

ID	Product	Sales	
1	Caps	10	
2	Caps	30	
3	Gloves	5	
4	Gloves	20	

ID	Product	Total Sales	
1	Caps	40	Σ
2	Caps	40	Σ
3	Gloves	25	Σ
4	Gloves	25	Σ

It **doesn't collapse** rows into a single row
(Row-Level-Calculations)

GROUP BY

Functions

Aggregate Functions

COUNT (*expr*)

SUM (*expr*)

MAX (*expr*)

MIN (*expr*)

MIN (*expr*)

WINDOW

Functions

Aggregate Functions

COUNT (*expr*)

SUM (*expr*)

MAX (*expr*)

MIN (*expr*)

MIN (*expr*)

Rank Functions

ROW_NUMBER ()

RANK ()

DENSE_RANK ()

CUME_DIST ()

PERCENT_RANK ()

NTILE (*n*)

Value (Analytics) Functions

LEAD (*expr, offset, default*)

LAG (*expr, offset, default*)

FIRST_VALUE (*expr*)

FIRST_VALUE (*expr*)

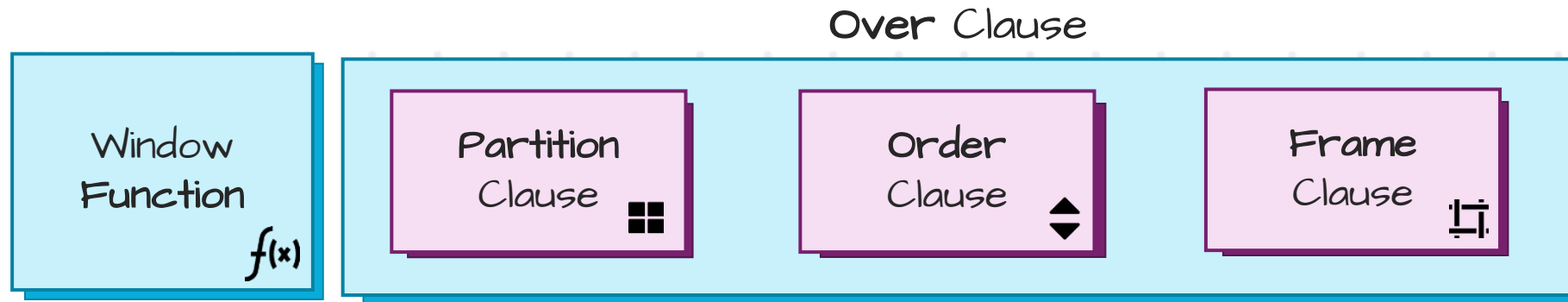
GROUP BY

**Simple Data Analysis
(Aggregations)**

WINDOW

**Advanced Data Analysis
(Aggregations + Details)**

Window Syntax



Window Syntax

```
AVG(Sales) OVER ( PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING )
```

Window Syntax

Calculation used
on the Window

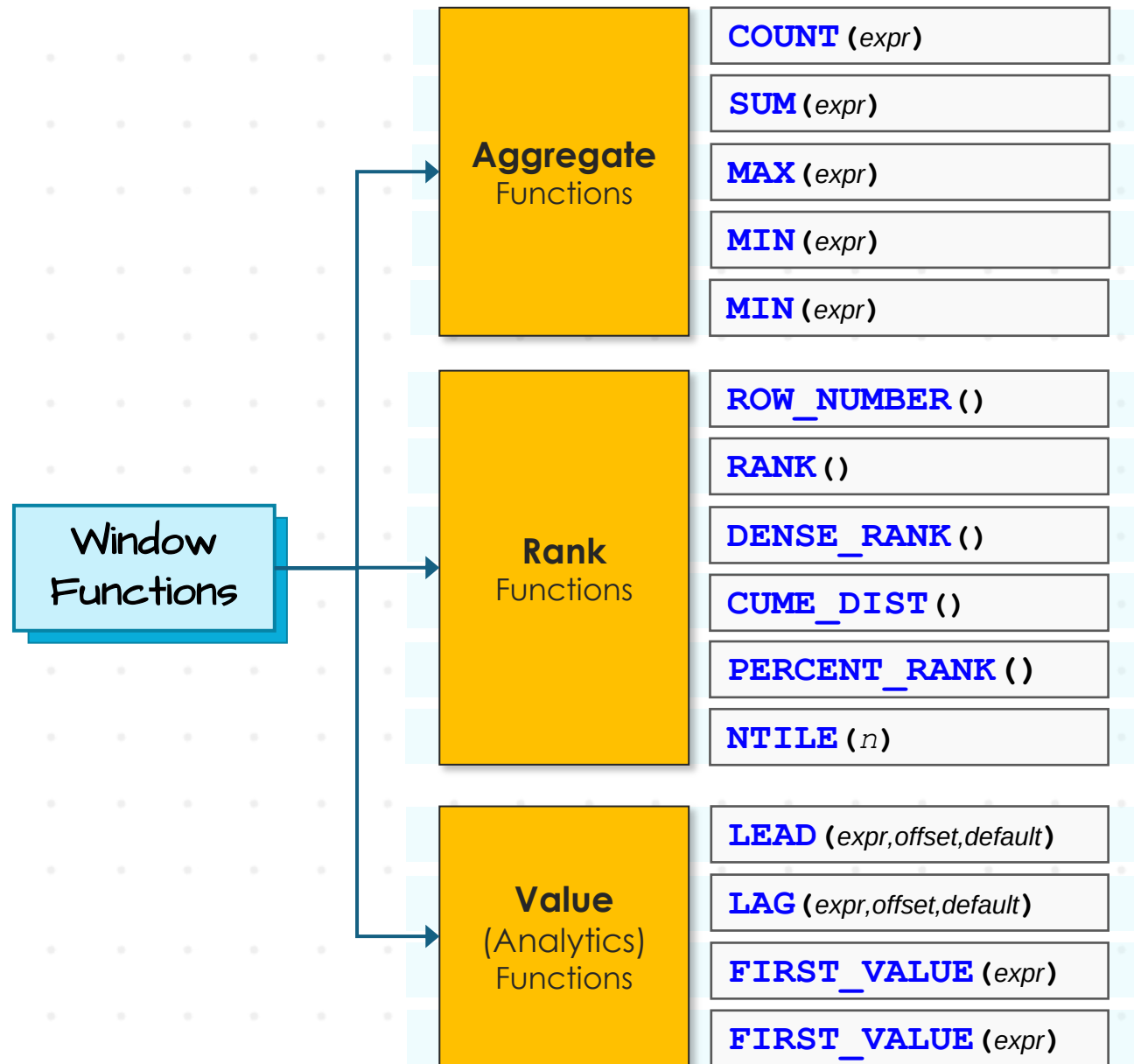
f(*) Window
Function

AVG (Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)



WINDOW FUNCTIONS

Perform calculations within a window



Window Syntax

Calculation used
on the Window

f(*) Window
Function

AVG (Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

Function
Expression

FUNCTION EXPRESSION

Arguments you pass to a function

Window Expressions

Empty

`RANK() OVER (ORDER BY OrderDate)`

Column

`AVG(Sales) OVER (ORDER BY OrderDate)`

Number

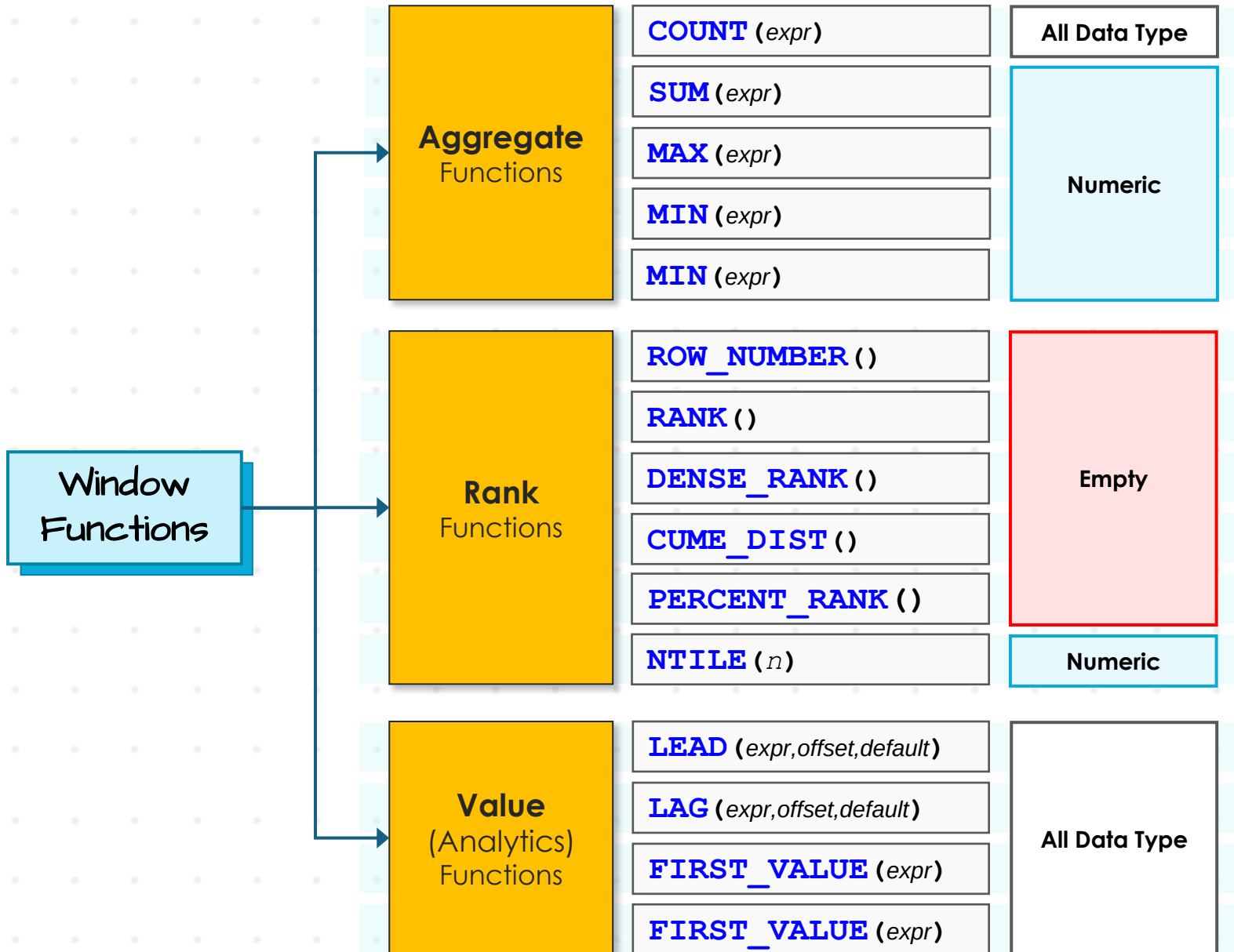
`NTEIL(2) OVER (ORDER BY OrderDate)`

Multiple Arguments

`LEAD(Sales,2,10) OVER (ORDER BY OrderDate)`

Conditional Logic

`SUM(CASE WHEN Sales > 100 THEN 1 ELSE 0 END) OVER (ORDER BY OrderDate)`



Window Syntax

Calculation used
on the Window

f(*) Window
Function

Define the
Window

Over
Clause

AVG (Sales)

OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

Function
Expression

OVER CLAUSE

Tells SQL that the function used is a **window function**

Window Syntax

Calculation used
on the Window

$f(*)$ Window
Function

Define the
Window

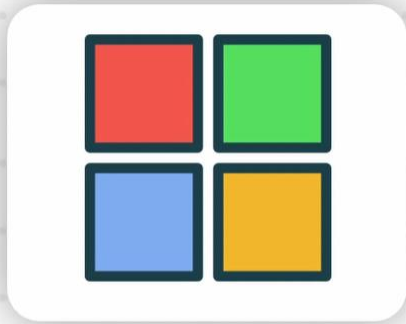
Over
Clause

AVG (Sales) **OVER** (**PARTITION BY** Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

Function
Expression

Partition
Clause 

Divides the dataset into
windows (Partitions)



PARTITION BY

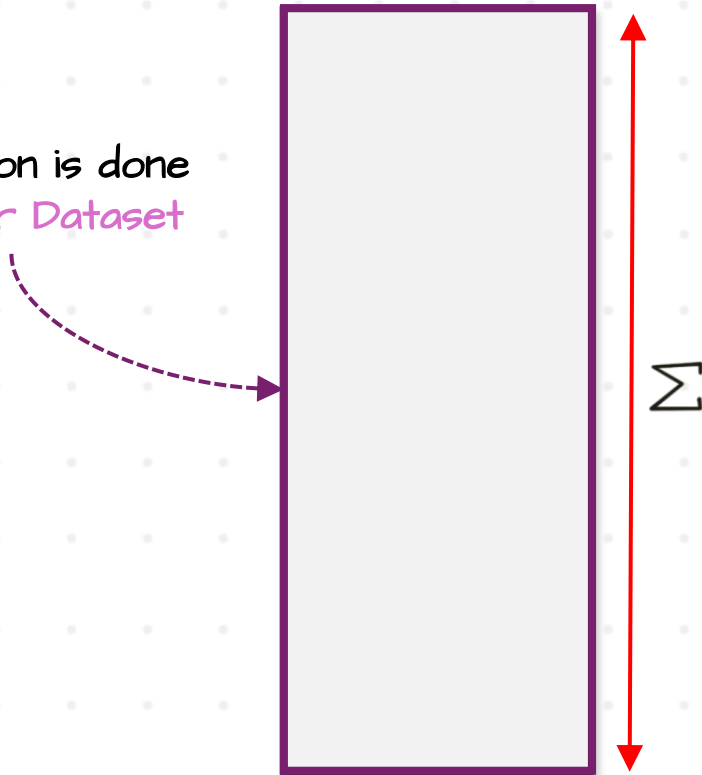
Divides the result set into partitions (Windows)

Partition By

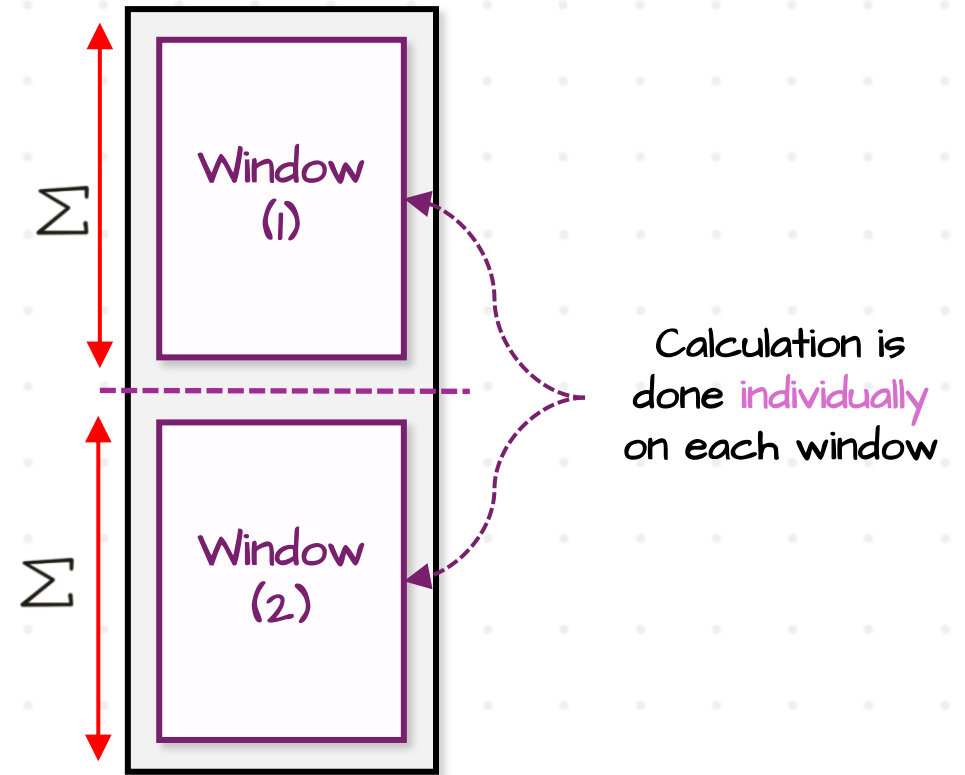
PARTITION BY divides the rows into groups, based on the column/s

`SUM(Sales) OVER ()`

Calculation is done
on *entier Dataset*



`SUM(Sales) OVER (PARTITION BY Product)`



Partition By

PARTITION BY divides the rows into groups, based on the column/s

Without
Partition By

Total sales across all rows (Entire Result Set)

```
SUM(Sales) OVER ()
```

Partition By
Single Column

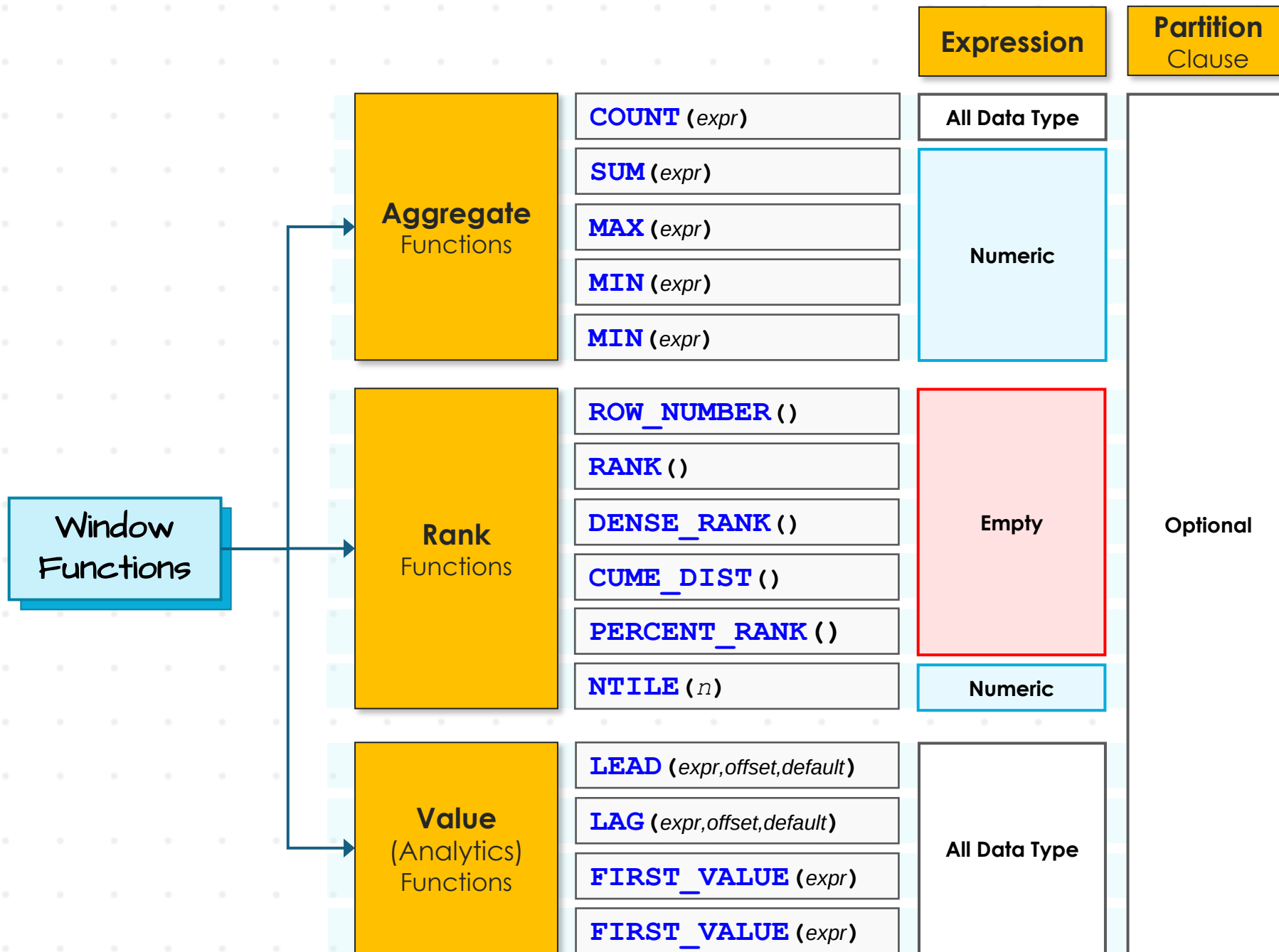
Total sales for each Product

```
SUM(Sales) OVER (PARTITION BY Product)
```

Partition By
Combined-Columns

Total sales for each combination of Product and Order Status

```
SUM(Sales) OVER (PARTITION BY Product, OrderStatus)
```



Window Syntax

Calculation used
on the Window

$f(*)$ Window
Function

Define the
Window

Over
Clause

AVG (Sales) **OVER** (**PARTITION BY** Category **ORDER BY** OrderDate **ROWS UNBOUNDED PRECEDING**)

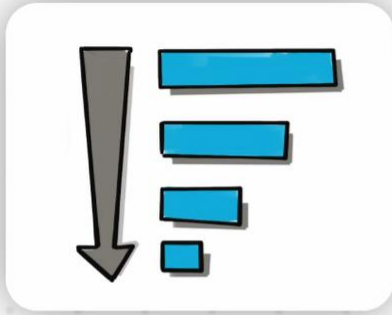
Function
Expression

Partition
Clause 

Divides the dataset into
windows (Partitions)

Order
Clause 

Sort the data
in a window



ORDER BY

Sort the data within a window

(Ascending | Descending)

		Expression	Partition Clause	Order Clause
Window Functions	Aggregate Functions	COUNT (expr)	All Data Type	Optional
		SUM (expr)		
		MAX (expr)		
		MIN (expr)		
		MIN (expr)		
	Rank Functions	ROW_NUMBER ()	Empty	Required
		RANK ()		
		DENSE_RANK ()		
		CUME_DIST ()		
		PERCENT_RANK ()		
		NTILE (n)		
	Value (Analytics) Functions	LEAD (expr,offset,default)	All Data Type	Required
		LAG (expr,offset,default)		
		FIRST_VALUE (expr)		
		FIRST_VALUE (expr)		

Window Syntax

Calculation used on the Window

$f(*)$ Window Function

Define the Window

Over Clause

AVG (Sales) OVER (PARTITION BY Category ORDER BY OrderDate ROWS UNBOUNDED PRECEDING)

Function Expression

Partition Clause 

Divides the dataset into windows (Partitions)

Order Clause 

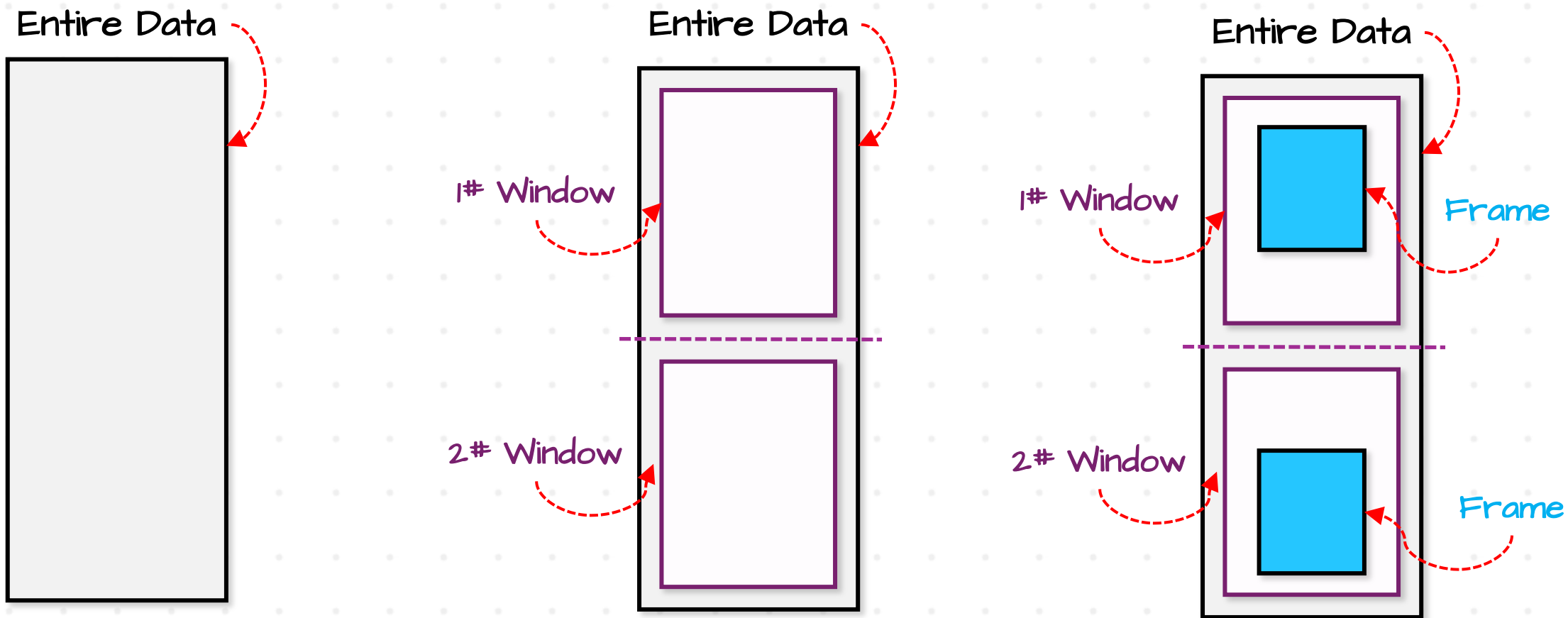
Sort the data in a window

Frame Clause 

Define a subset of rows in a window



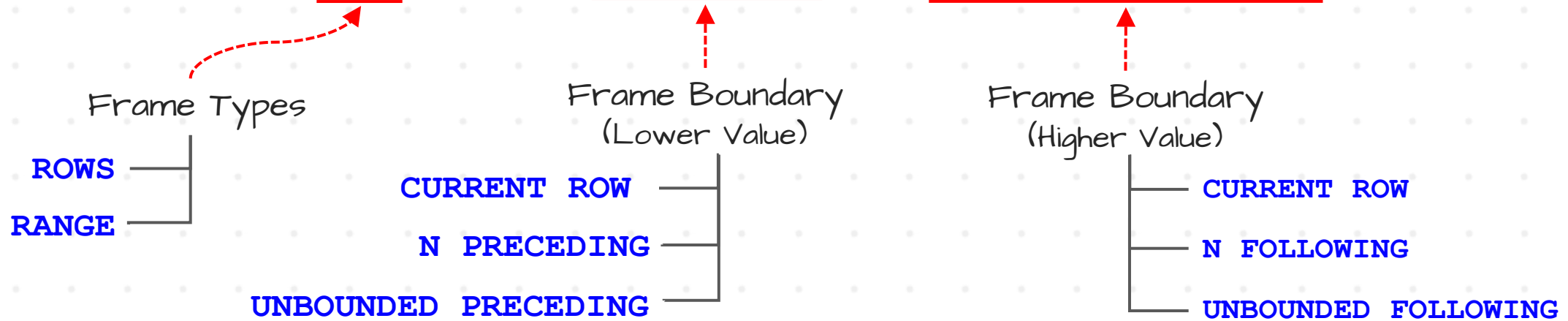
Frame Clause



		Expression	Partition Clause	Order Clause	Frame Clause
Window Functions	Aggregate Functions	COUNT (expr) SUM (expr) AVG (expr) MAX (expr) MIN (expr) Numeric	Optional	Optional	Optional
	Rank Functions	ROW_NUMBER () RANK () DENSE_RANK () CUME_DIST () PERCENT_RANK () NTILE (n) Empty Numeric		Required	Not allowed
	Value (Analytics) Functions	LEAD (expr,offset,default) LAG (expr,offset,default) FIRST_VALUE (expr) FIRST_VALUE (expr) All Data Type		Required	Not allowed Optional Should be used

AVG(Sales) OVER (PARTITION BY Category ORDER BY OrderDate

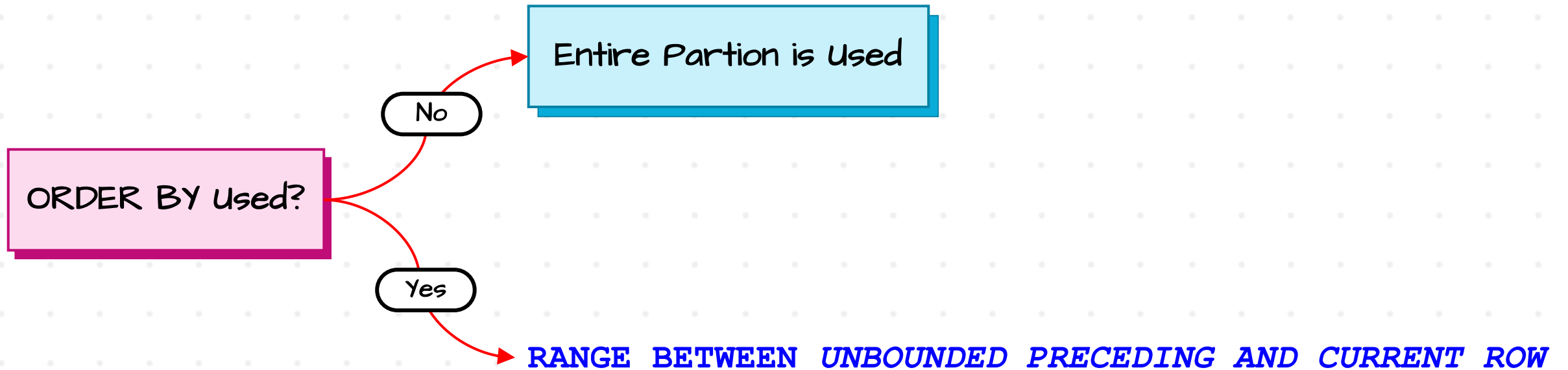
ROWS BETWEEN CURRENT ROW AND UNBOUNDED FOLLOWING)



Rules

- Frame Clause can only be used **together** with order by clause.
- Lower Value must be **BEFORE** the higher Value.

Frame



COMPACT FRAME

For only PRECEDING, the CURRENT ROW can be skipped

NORMAL FORM

ROWS BETWEEN CURRENT ROW AND 2 FOLLOWING

SHORT FORM

ROWS 2 FOLLOWING

Window Rules

#1 RULE

Window functions can be used ONLY
in **SELECT** and **ORDER BY** Clauses

#2 RULE

Nesting Window Functions is not allowed !

#3 RULE

SQL **execute** WINDOW Functions **after** WHERE Clause

#4 RULE

Window Function can be used together with GROUP BY
in the same query, ONLY if the same **columns** are used

Window Rules

Inner Window Function

AVG (SUM(Sales) OVER ()) OVER (ORDER BY OrderDate)

Outer Window Function



Not allowed to nest
window functions !

Windowed functions cannot be used in the context of another windowed function or aggregate.

SQL WINDOW FUNCTIONS

Performs calculations on subset of data without losing details

Window v/s Group By

- Window is more powerfull & Dynamic than Group By.
- Data Analysis
 - Advanced → Window
 - Simple → Group By
- Use Group By + window in same Query, only if same Column used.

Components

Window Functions + Window Definition **OVER**



Rules

- Nesting is not allowed!
- Window Can be used only in **SELECT** and **ORDER BY**
- SQL executes window **AFTER** filtering data using **WHERE**

$f(x)$ Window Functions

Aggregate

- SUM ()
- AVG ()
- COUNT ()
- MAX ()
- MIN ()

Perform calculations on a set of rows and return a **single aggregated** value for each row

Rank

- ROW_NUMBER ()
- RANK ()
- DENSE_RANK ()
- NTILE ()
- CUME_DIST ()
- PERCENT_RANK ()

Assign a **rank** to each row in a window

Value

- LAG ()
- LEAD ()
- FIRST_VALUE ()
- LAST_VALUE ()

Return a **specific value** in a window to be compared with the value of **current row**